

Issues Found in the Bad Version

1. DRY (Don't Repeat Yourself)

- a. The *player_x_turn()* and *player_o_turn()* functions contain nearly identical code.
- b. The *check_stuff()* function repeats the winner-checking logic for every possible X and O win condition.

2. Poor Variable Names

- a. Variables *a* through *I* are used to represent board spaces.
- b. These names do not clearly describe their purpose, making the code harder to read.

3. Overuse of Global Variables

- a. The *global* keyword is used throughout the program.
- b. This makes it difficult to track where data is being modified.

4. Single Responsibility Principle (SRP)

- a. The *player turn* functions handle multiple tasks: user input, validation, and board updates.
- b. The *main* program manually controls each turn instead of using a loop.

5. YAGNI (You Aren't Gonna Need It)

- a. The *useless_future_features()* function was included even though it is never used.
- b. It adds unnecessary code for features that do not exist.

Improvements in the Good Version:

1. Replaced nine separate variables with a single board list.
2. Combined player move logic into one reusable function.
3. Used a loop to manage turns instead of repeating code.
4. Stored winning combinations in a list to eliminate repeated if statements.
5. Removed unnecessary future-feature code.